

NAMAN SINGH

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PROFESSIONAL SUMMARY

- B.Tech Computer Science Engineering (Information Technology / AI & ML specialization) undergraduate at VIT Bhopal University with strong technical knowledge across Python, Java, and Artificial Intelligence / Machine Learning, and a track record of taking projects from prototype to working, deployable software.
- Solid foundation in Data Structures, Algorithms, and Object Oriented Programming (OOP), with 109+ problems solved on LeetCode and active competitive programming on Codeforces — strong problem solving and debugging ability under time constraints.
- Hands-on Software Engineering experience building REST APIs and backend services (Flask, FastAPI) connecting ML/Deep Learning models to real, client-facing applications — comfortable owning a feature end-to-end from data to deployment.
- Created Generative AI / NLP systems including a Retrieval-Augmented Generation (RAG) pipeline with hallucination detection, plus Computer Vision systems for real-world detection, using Python, TensorFlow, and PyTorch.
- Strong communication and cross-functional collaboration skills; comfortable with rapid prototyping in fast-paced, client-driven settings — suited to a Forward Deployed Engineer role blending engineering with applied AI/ML.

RELEVANT PROJECTS

ContextRAG — End-to-End RAG Pipeline with Hallucination Detection *Python, FAISS, ChromaDB, Sentence-Transformers, Ollama, Groq*

- Engineered a retrieval-augmented generation (NLP) pipeline with 2 swappable LLM backends (local Ollama, cloud Groq) across a 4-stage architecture (ingest, chunk, embed, retrieve), enabling flexible, production-style deployment.
- Implemented dense vector retrieval across 2 vector database backends (FAISS, ChromaDB) with sentence-transformer embeddings, applying object-oriented design to keep retrieval and generation components modular, testable, and independently deployable.
- Designed a hallucination detection layer that flags low-confidence or unsupported model outputs before they reach the user, improving response reliability for client-facing use cases.
- Documented the full architecture, experiments, and evaluation results across all 4 pipeline stages, cutting onboarding time for anyone extending the project.

Product Authenticity Analysis System

Python, TensorFlow, Flask, OpenCV, HTML/CSS/JS

- Developed a multi-modal Machine Learning and Computer Vision system for product authenticity detection, fusing 4 signal types — MobileNetV2 image classification, OpenCV visual checks, pricing analytics, and metadata processing — into one scoring pipeline.
- Analyzed 500,000+ product listings with Pandas and NumPy to build a pricing-anomaly detection engine, applying core data structures and algorithms to keep large-scale processing efficient.
- Integrated 4 independent risk signals — visual, pricing, vendor-trust, and text-risk — through a Flask backend REST API, following software engineering best practices for modular, testable code.
- Delivered a production-oriented architecture with asynchronous uploads and real-time progress tracking, deployed as a working, client-facing web application.

Smart AI Resume Analyzer & SaaS Platform

Python, Streamlit, SQLite, Gemini API, Selenium

- Architected a SaaS-style Artificial Intelligence resume analyzer using the Gemini API to evaluate ATS compatibility and generate personalized improvement recommendations.
- Achieved 98%+ text-extraction precision across scanned and digital resumes by combining 3 parsing techniques (PyPDF2, DOCX processing, Tesseract OCR) with rigorous testing.
- Automated a real-time job recommendation engine aggregating listings from 2 platforms (LinkedIn, Naukri) via Selenium web scraping, with SQLite-backed analytics and feedback storage.

Reinforcement Learning: Q-Learning GridWorld

Python, NumPy, Matplotlib

- Trained a Q-Learning agent in a custom 5x5 GridWorld environment across 3 exploration strategies, analyzing convergence speed and reward stability to study AI decision-making.

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL — strong Data Structures, Algorithms, and Object-Oriented Programming (OOP) foundations

AI / ML / Deep Learning: Scikit-learn, TensorFlow, PyTorch, NumPy, Pandas — Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision

Generative AI: RAG, LangChain, OpenAI API, Prompt Engineering, FAISS, ChromaDB, Sentence-Transformers

Software Engineering & Backend: FastAPI, Flask, REST APIs, SQLite, deployment, debugging, testing, cloud-ready architecture

Tools & Practices: Git, GitHub, VS Code, Jupyter Notebook, cross-functional collaboration, client communication

EDUCATION

VIT Bhopal University — B.Tech, Computer Science Engineering (AI & ML Specialization) *Sep 2023 – Present*

- CGPA: 8.42 | Coursework: ML, Deep Learning, Data Structures & Algorithms, NLP, Reinforcement Learning

Holy Cross School, Bengaluru — Secondary Education

Jun 2021 – Mar 2023

- Class X: 95% | Class XII: 85%